

DISHAN SHAIKH

AI & ML Engineer | Full Stack Developer | Data Science | computer vision

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Portfolio- [Dishan Shaikh | AI & ML Engineer — Portfolio](#), Linkedin profile- [linkedin.com/in/dishan-shaikh-68b116311](#),

B.Tech CSE (AI & ML)

PCCOE, Pune | 2024–2028

3rd Year Student

Open to Internships & Projects

PROFESSIONAL SUMMARY

Results-driven AI & ML Engineering student (Pre-final year, B.Tech CSE–AIML, PCCOE) with deep expertise in **Machine Learning, Generative AI, Data Science, LLMs, Agentic AI, Computer Vision, and Full Stack Development**. Proven track record of building end-to-end AI system, Passionate about deploying scalable, intelligent systems that solve real-world sustainability and governance challenges.

TECHNICAL SKILLS

AI / ML: Machine Learning · Deep Learning (CNN, Autoencoders, Transformers) · Generative AI · LLMs (GPT-4, Claude, Gemini) · Agentic AI · RAG Pipelines · Computer Vision (YOLOv8/v10) · NLP · Prompt Engineering

Frameworks & Libraries: TensorFlow · PyTorch · Keras · Scikit-learn · OpenCV · LangChain · LlamaIndex · HuggingFace · Pandas · NumPy · Matplotlib · Seaborn

Data Science: Data Cleaning · Feature Engineering · EDA · Statistical Analysis · Data Visualization · Environmental Data Modeling · A/B Testing

Programming Languages: Python · C++ · Kotlin · JavaScript (ES6+) · TypeScript · HTML5 · CSS3 · SQL

Full Stack Development: Node.js · Express.js · React.js · Next.js · REST APIs · GraphQL · JWT Auth · MongoDB · PostgreSQL · Firebase · Supabase

Cloud & DevOps: AWS (EC2, S3) · Google Cloud Platform · Docker · Git & GitHub · CI/CD Pipelines · Vercel · Railway

Tools & Platforms: VS Code · Jupyter Notebook · Google Colab · Postman · Figma · Arduino IDE · Notion

PROJECTS

Elevate – AI-Powered Learning Platform | React · Node.js · LLMs · RAG · LangChain 2026–Present

- Designed and built a full-stack adaptive learning platform that personalizes content delivery using LLM-driven RAG pipelines
- Integrated AI tutor with multi-turn conversational memory, auto-quiz generation, and concept gap detection
- Built progress analytics dashboard with learning velocity tracking and spaced-repetition scheduling
- Implemented JWT-secured auth, role-based access (student/instructor), and course builder with rich-text editor

React · Node.js · MongoDB · LangChain · OpenAI API · JWT · Tailwind CSS

AI Avatar – Government Citizen Services Platform | Python · LLMs · Computer Vision · TTS · FastAPI 2026–Present

- Engineered a lifelike AI avatar system for government portals enabling citizens to interact via natural language in Hindi, Marathi & English
- Integrated multi-modal pipeline: speech-to-text → LLM reasoning → text-to-speech → lip-synced avatar video rendering
- Deployed on-premise with offline-capable fallback for low-bandwidth govt environments
- Reduced citizen query resolution time by ~65% compared to traditional IVR systems in pilot testing

Python · FastAPI · Whisper STT · LLM APIs · D-ID / SadTalker · Multilingual NLP

Advanced AQI Prediction System | Python · ML · CNN · Environmental Modeling 2024

- Built ML model integrating real-time traffic density and environmental sensor data for hyperlocal air quality forecasting
- Implemented CNN-based spatial pollution distribution modelling achieving 91% prediction accuracy
- Designed alert system with auto-notification triggers for hazardous AQI thresholds

Python · TensorFlow · CNN · Scikit-learn · Pandas · OpenWeatherMap API

Vehicle-wise Pollution Estimation using YOLOv8 | Python · Computer Vision · YOLO 2025

- Developed real-time computer vision pipeline for multi-class vehicle detection and emission factor classification
- Estimated per-vehicle emission patterns by fusing YOLO detections with govt pollution norms database

▶ Achieved 87% mAP on custom dataset of 5,000+ annotated vehicle images
YOLOv8 · OpenCV · Python · Roboflow · Custom Dataset Annotation

Carbon Emission Tracking & Accountability Platform | Full Stack · Predictive Analytics 2026

- ▶ Designed microservices architecture for real-time carbon emission tracking across industrial categories
- ▶ Integrated predictive analytics engine with compliance scoring and regulatory reporting dashboards
- ▶ Built REST API consumed by 3 pilot organizations for sustainability benchmarking

Node.js · React · MongoDB · Python ML · REST APIs · Chart.js

Solar Powered Car | Arduino · Embedded Systems · Team Project 2024

- ▶ Led embedded systems sub-team; designed power management circuit for solar-to-motor energy conversion
- ▶ Programmed Arduino-based telemetry system for real-time battery, speed and solar output monitoring

Arduino · C++ · Embedded Systems · PCB Design · PCCOE–NSS

EDUCATION

B.Tech in Computer Science & Engineering (AI & ML) 2024 – 2028

Pimpri Chinchwad College of Engineering (PCCOE), Pune

Current Year: 3rd Year | Branch- CSE(AI & ML), Specialization: Artificial Intelligence & Machine Learning

CERTIFICATIONS & ACHIEVEMENTS

- ▶ Oracle cloud infrastructure generative AI professional course
- ▶ Finalist – INDIA INNOVATES DELHI (offline) Hackathon 2026 (AI Avatar project)
- ▶ participated in Convolve 4.0 - A Pan-IIT AI/ML organised by Indian Institute of Technology IIT, Guwahati
- ▶ Final round- Indradhanu hackathon -(AI For climate change)

SOFT SKILLS & LANGUAGES

Soft Skills: Leadership · Critical Thinking · Research Mindset Problem Solving · Team Collaboration · Communication Agile/Scrum Familiarity · Fast Learner · Ownership Mindset	Languages: English (Professional) Hindi (Native) Marathi (Native)
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